

Task - 12: Log Monitoring & Analysis

Internship Task – Elevate Labs

This task focused on understanding the importance of system and security logs and how log monitoring helps in detecting security incidents. The objective was to analyze logs, identify suspicious activity, and understand the basics of incident detection.

1. Introduction to Log Monitoring

Logs are records of events generated by operating systems, applications, and network devices. They provide valuable information about system activities and user behavior.

2. Types of Logs

Different types of logs were studied, including: - Authentication logs - System logs - Application logs - Security logs

Each log type helps in identifying specific events and issues.

3. Authentication Log Analysis

Authentication logs were analyzed to monitor user login activities. Failed login attempts were identified to detect possible brute-force or unauthorized access attempts.

4. Identifying Failed Logins

Multiple failed login attempts from the same user or IP address were considered suspicious. Such patterns can indicate password attacks.

5. Anomaly Detection

Anomalies such as unusual login times or unexpected access attempts were identified. Detecting anomalies helps in early identification of security incidents.

6. Event Correlation

Different log events were correlated to understand attack patterns. For example, multiple failed logins followed by a successful login can indicate compromise.

7. Introduction to SIEM

Security Information and Event Management (SIEM) systems collect and analyze logs from multiple sources. SIEM tools help automate detection and alert generation.

8. Writing Alerts

Basic alert conditions were defined for suspicious activities such as repeated failed logins. Alerts help security teams respond quickly to incidents.

9. Documentation of Findings

All observations and detected issues were documented clearly. Proper documentation supports incident response and future analysis.

Learning Outcome

Through this task, I gained practical knowledge of log monitoring and analysis. I learned how logs help detect security incidents and how basic SIEM concepts support effective incident detection.